

Population dynamics and sustainability of the spiny lobster (*Panulirus meripurpuratus* Giraldez & Smyth, 2016) fishery on the Amazon continental shelf

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Abstract. The assessment of fishery resources is key to the development of management strategies for exploited stocks. Analysis of specific parameters provides information regarding the state of the stock and indicators of exploitation levels. The aim of the present study was to analyse the growth and mortality parameters, as well as investigate the exploitation status, of *Panulirus meripurpuratus* caught on the Amazon coast, northern Brazil. Growth parameters and stock assessment were estimated using R with bootstrapping and FISAT II, based on cephalothorax length–distribution frequency. Specimens were collected by the commercial lobster fishery off the coast of the states of Pará and Amapá between 2001 and 2002. The results indicate that the *P. meripurpuratus* stock assessed is composed of individuals with low growth rates and high estimates of maximum asymptotic length, with relative longevity of 21.37 years. The exploitation rate was operating at near the maximum limit of sustainability in 2001 and 2002. This is the first estimate of the exploitation status of *P. meripurpuratus* on the Amazon coast. We strongly recommend new monitoring efforts and further studies to support the conservation and management of the spiny lobster stock.

Additional keywords: fishery management, growth and mortality parameters, stock assessment.

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Introduction

The Palinuridae spiny lobsters are long-lived benthic crustaceans that exhibit migratory and gregarious behaviour (George 2005; Cruz *et al.* 2007). They are large-bodied omnivorous crustaceans with a nocturnal habit, preferentially carnivorous and represent a high-value fishery resource in several regions around the world (Fonteles-Filho 2005; Phillips *et al.* 2013). These crustaceans have late sexual maturity, fragmented spawning and indirect development (up to 24 months) through different larval stages before the juvenile phase (Phillips *et al.* 2013; Baeza *et al.* 2016). In tropical and temperate regions,

lobsters are captured by industrial and artisanal fleets using lobster traps, bottom-set gill nets and trawls, or manually by free divers or divers using compressors and either gaffs or harpoons (Conrad and Danoff-Burg 2011; Penn *et al.* 2015).

Three species (*Homarus americanus*, *Nephrops norvegicus* and *Panulirus argus*) from two families (Nephropidae and Palinuridae) are responsible for more than 80% of the lobster biomass landed worldwide (Ward and Phillips 2013). The Family Palinuridae accounts for the majority of the lobster catches in countries such as South Africa, Australia, New Zealand, Cuba, the US, Mexico, The Bahamas and Brazil (Bowen 1980;

Andrade 2015). There are at least two populations of *Panulirus* in the western Atlantic, one located in the US and Caribbean (*Panulirus argus*) and another on the Brazilian coast (*Panulirus meripurpuratus*; Giraldes and Smyth 2016). In 2017, the mean global catch of lobsters was 309 739 tonnes (Mg); in the western Atlantic, *P. argus* was the main species harvested, at 39 236 Mg (Food and Agriculture Organization of the United Nations 2019; see also <http://www.fao.org/fishery/>, accessed 21 January 2020), which represents 12.66% of total production worldwide.

The lobster is one of the main exports of Brazilian fisheries, with production reaching ~6929.2 Mg in 2011. The US is the main consumer market for Brazilian lobster, bought 2134.773 Mg, which represents a cost of US\$69 163 978, making it (Ministério da Pesca e Aquicultura 2011). *P. meripurpuratus* is the main lobster caught along the Brazilian Amazon coast; in 2007 (the last year that official Brazilian fishery statistics were available for all states), the state of Pará produced 1355.5 Mg, generating a revenue of US\$17 716 613.02 (Instituto Brasileiro do Meio Ambiente e dos Recursos Naturais Renováveis 2007), calculated on the basis of the mean annual exchange rate for the US dollar in 2007, corrected by the General Price Index – Internal Availability (GPI-IA) of the Getúlio Vargas Foundation.

The first artisanal catches of lobster were landed in Brazil at the end of the 1950s in Pernambuco state (Paiva 1997). Lobster fishing intensified over the following decades, expanding along the entire north-east coast of Brazil and becoming totally or partially industrialised (Paiva 1997). This resulted in higher exploitation rates and a subsequent decline in the region's stocks (Cruz *et al.* 2013a). Faced with declining stocks, operations expanded to more productive fishing grounds to the north and south of the originally exploited area (Porto *et al.* 2005). Currently, commercial lobster fisheries operate along most of the Brazilian coast, between the states of Amapá and Rio de Janeiro (Tubino *et al.* 2007). According to Instituto Brasileiro do Meio Ambiente e dos Recursos Naturais Renováveis (2007), Ceará is the largest producer, followed by Pará state.

Lobster fisheries started off the Brazilian Amazon coast in the 1970s, intensifying in the 1990s and peaking in 2004 (Instituto Brasileiro do Meio Ambiente e dos Recursos Naturais Renováveis 2005). Despite the growing demand for Brazilian lobster and the increase in international market prices, catches have declined since this peak (Food and Agriculture Organization of the United Nations 2003; Instituto Brasileiro do Meio Ambiente e dos Recursos Naturais Renováveis 2005, 2007). A similar decline in catches was reported along the east coast of the US and well as in Cuba, the Caribbean and other Brazilian regions. In general, the reduction in productivity of lobster fisheries has been attributed to factors such as changes in the environmental conditions in areas where larvae develop associated with extreme climatic events (e.g. El Niño), habitat loss and intense fishery pressure on the resource (Arce and de León 2001; Puga *et al.* 2013). Another factor that could affect the fishing resource off the Amazon coast is the change in ocean variables related to the high variability that the Amazon hydrological cycle has been exhibiting in recent decades (since 1990; Gouveia *et al.* 2019a). An additional problem is the 'ghost fishery', specifically fishing gear lost at sea that may be responsible for undesirable high mortality rates (Butler and Matthews 2015).

The exploitation of lobster stocks along the Amazon coast is considered one of the least sustainable fishery activities (Isaac *et al.* 2009). Fishing off the northern Brazilian coast primarily uses bottom gill nets, which capture individuals of all sizes and so can modify the structure of the substrate and benthic fauna, which is often caught as bycatch (Instituto Brasileiro do Meio Ambiente e dos Recursos Naturais Renováveis 2008).

Estimates of growth parameters can provide essential insights into fishery resource dynamics and can be used as input for fisheries assessment models (Gulland 1955). However, few data are available for the fishery population parameters of Brazilian lobster stocks. The studies that are available were performed in the 1970s, 1980s and 1990s, focusing primarily on the features of the lobster fishing fleets in Ceará and Pernambuco states (Fonteles-Filho 1979, 1986, 1997; Vasconcelos *et al.* 1994; Paiva 1997; Santos and Freitas 2002; Andrade 2015; Oliveira *et al.* 2015), which exploit fishing grounds far from the Amazon coast. By contrast, data from northern Brazil are limited to a description of fishery techniques and characterisation of fishery activities (Porto *et al.* 2005; Santos *et al.* 2019), as well as studies of reproductive patterns and larval recruitment (Silva *et al.* 2008; Cruz *et al.* 2014, 2015; Lima *et al.* 2018) and the composition and size of species caught (Silva *et al.* 2003, 2007, 2013; Cruz *et al.* 2013a, 2013b).

Recent research indicates that the north of the Amazon River mouth is characterised as a differentiated biogeographic province due to the interaction between continental and oceanic waters, which affects the variability of both primary productivity and dissolved organic carbon, local fish communities (Medeiros *et al.* 2015; Andrades *et al.* 2018; Di Paolo *et al.* 2019; Gouveia *et al.* 2019b) and, consequently, lobsters. The retroflexion of the North Brazil Current and Amazon–Orinoco plume, even though permeable, acts as a barrier to the dispersal of lobster larvae and separates the Brazilian (*P. meripurpuratus*) and Caribbean (*P. argus*) lobster stocks (Giraldes and Smyth 2016; Cruz *et al.* 2015). These factors, combined with the different exploitation patterns within the two regions along the Brazilian coast (i.e. northern and north-east), support the hypothesis that there are two different stocks with different population parameters and conservation status.

A new methodology of determining growth patterns based on the rigid structures of a crustacean (i.e. gastric ossicles or eyestalk) is now available (Kilada *et al.* 2012; Gnanalingam *et al.* 2019). However, the most common methodology to determine growth patterns is through either capture–mark–recapture studies (e.g. Ehrhardt 2008; Zhang and Yao 2018) or the analysis of body length distribution (e.g. Skewes *et al.* 1997; de León *et al.* 2005). Given the intrinsic methodological difficulties and the limited resources available for scientific research in developing countries, no mark–recapture and rigid structure studies of *P. meripurpuratus* are currently available in northern Brazil. This reinforces the applicability of body length analyses for the investigation of growth patterns, especially because this approach is simple and inexpensive.

In the past 40 years, fishing effort has increased but lobster production has decreased by an average of 30%; nowadays *P. meripurpuratus* is classified as Near Threatened (NT) in the Brazilian crustacean Red List (Santana *et al.* 2016). Because the current status of lobster exploitation along the Amazon coast is

unknown, the aim of the present study was to analyse the growth and mortality parameters and to assess the state of lobster stocks for this region. The findings of this study provide good insights for the development of effective stock management strategies along the Amazon coast and important information for future comparisons.

Materials and methods

Study area

The *P. meripurpuratus* fishing grounds on the Amazon coastal shelf are located between 01°30'N and 02°20'N in Pará state and between 02°30'N and 03°50'N in Amapá state (Fig. 1). The seabed of this region is characterised by different substrate types, although sand, gravel and coral dominate the lobster fishing areas (Silva *et al.* 2008).

This region is affected by significant input of river water, particularly from the Amazon and Tocantins rivers, which reach their maximum discharge during the first half of the year (rainy season). During this season, when heavy rains flood the local rivers, and a freshwater plume is created that can extend 350 km from the mouth of the Amazon River into the Atlantic Ocean (Barthem and Schwassmann 1994). During the second half of the year, there is a marked decrease in rain and a corresponding decrease in the extent of the plume, allowing the saltier water of the ocean to advance towards the coast, increasing the salinity of coastal areas (Barthem and Schwassmann 1994).

In addition, this area is affected by the North Brazil Current, which transports the warmest tropical waters to the South Atlantic (Sanyo Techno Marine 1998; Silva *et al.* 2009). Although this region is highly productive and has a rich and diversified biota, the marine environment is considered stratified and oligotrophic, with nutrients being derived primarily from either the transportation of organic matter from local mangrove forests or the Amazon River plume, because primary productivity is limited by the turbidity of the water, which reduces the potential for the penetration of sunlight (Isaac and Ferrari 2017).

Data collection

The data analysed herein were obtained from the Lobster Project, coordinated by the Centro de Pesquisa e Gestão de Recursos Pesqueiros do Litoral Norte (CEPNOR) at the Brazilian Institute for the Environment and Renewable Natural Resources (Instituto Brasileiro do Meio Ambiente e dos Recursos Naturais Renováveis, IBAMA). This project compiled data on lobster catches by on-board monitoring of the commercial fleet, which operates using bottom-set gill nets and traps at depths of 60–100 m. During this monitoring, random samples of *P. meripurpuratus* individuals were collected between July and December 2001 and in May, July, August, October, November and December 2002. Individuals were identified by morphological characteristics; their cephalothorax length (CL), abdomen length (AL) and total length (TL) were measured in millimetres, and sex was determined based on the observation of secondary external characteristics (Dall'occo 2010).

Data analysis

The non-parametric Kruskal–Wallis procedure was used to test the significance of differences in the mean lengths for both sexes. Biometric relationships were tested by applying linear

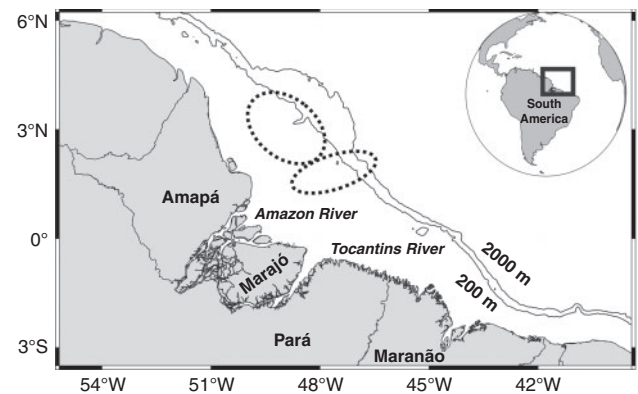


Fig. 1. Study area showing the fishing grounds where spiny lobster (*Panulirus meripurpuratus*) data were collected on the Amazon north coast (dotted lines). Black lines in the oceanic portion represent the isobaths of 200 and 2000 m.

regression to the dataset for each sex separately, and then to the full dataset. The slope value (b) of each regression reflects the type of relationship between a specimen's length and other biometric parameters (i.e. isometric or allometric). The significance of the biometric relationships was evaluated using analysis of variance (ANOVA).

The ELEFAN_GA_Boot function (see https://rdrr.io/github/rschwamborn/fishboot/man/ELEFAN_GA_boot.html; Schwamborn *et al.* 2019) was used to determine the asymptotic length (L_{∞}), growth constant (k) and the growth performance index (Φ'), as well as 95% confidence intervals (CIs), for the male, female and pooled dataset (Schwamborn *et al.* 2019). This ELEFAN-based fit method considers: (1) sample size and structure; (2) variability in the population; and (3) the sampling uncertainty of the individuals, considering sample size and the precision of the method. Thus, a full bootstrap was conducted on 10 000 random resamples in the R environment (R Foundation for Statistical Computing, Vienna, Austria). The output consists of a Kernel density distribution plot and parameter estimates with respective CIs. CLs were grouped in 2-mm classes to adjust the somatic growth for the complete dataset. This analysis was modified from the original model proposed by von Bertalanffy (1938) to remove theoretical age at length 0 (t_0) as follows:

$$L_t = L_{\infty} \left(1 - e^{-k(t)} \right)$$

where L_t is the CL (mm) at age t (years), L_{∞} is the asymptotic length (mm) and k is the growth constant (year^{-1}).

The L_{∞} and k values of the parameters obtained using the ELEFAN_GA_Boot function were used to create a length–age key, which was compared with data available for other *Panulirus* species given that other types of validation were not applicable because of the research conditions. In addition, the values obtained were used to calculate total mortality (Z) using the linearised capture curve, as suggested by Ricker (1975). The empirical equation of Pauly (1980) was used to calculate natural mortality (M):

$$\log(M) = -0.0066 - 0.279 \times \log(L_{\infty}) + 0.6543 \times \log(k) + 0.463 \times \log(T)$$

Table 1. Number of individuals, mean and range of cephalothorax length (CL) and abdomen length (AL) of all *Panulirus meripurpuratus* collected on the Amazon coast in 2001 and 2002, and for males and females separately

	Number of individuals	CL (mm)		AL (mm)	
		Range	Mean $\pm$ s.d.	Range	Mean $\pm$ s.d.
Pooled dataset	6933	65–174	100.98 $\pm$ 12.98	56–313	177.91 $\pm$ 18.88
Males	3236	65–174	104.50 $\pm$ 14.21	56–269	173.82 $\pm$ 17.98
Females	3697	68–144	97.90 $\pm$ 10.02	78–313	181.49 $\pm$ 18.92

where T is the mean annual water temperature, which was fixed at 26°C (Silva *et al.* 2005).

The fishery mortality rate (F) and the exploitation rate (E) were calculated from the estimates of Z and M using the following equations:

$$F = Z - M$$

$$E = F \div Z$$

Longevity ($A_{0.95}$) was estimated using the formula of Taylor (1958):

$$A_{0.95} = (t_0 + 2.996) \div k$$

Beverton and Holt's (1957) yield per recruit model was used to evaluate the stock exploitation status, under the 'knife-edge' selectivity assumption and the length at which 50% of individuals had a chance of being captured (L_{50}) were run in the FISAT II – FAO-ICLARM Stock Assessment Tool (FAO Fisheries and Aquaculture Department, see <http://www.fao.org/fishery/topic/16072/en>). This method estimates the exploitation rate, obtained with the maximum yield ($E_{\max}$) and at $E_{0.10}$, which represents the exploitation rate when the slope of the yield curve is 10% of the initial curve. This is considered the most conservative and reliable estimate to guarantee the sustainability of a fishery (King 2007). These values were compared with the current exploitation rate (E_{actual}) estimated for the species to diagnose the species' exploitation status.

Results

In all, 6933 specimens were analysed (53.32% females, 46.67% males; sex ratio 1 : 1.14). The mean $\pm$ s.d. CL of all individuals was 100.98 $\pm$ 12.98 mm, whereas mean AL was 313 $\pm$ 19 mm (Table 1). Mean CL was significantly higher in males than females (Kruskal–Wallis, $H_1 = 281.8174$, $n = 6933$, $P < 0.05$), whereas AL was significantly higher in females than males (Kruskal–Wallis, $H_1 = 411.9772$, $n = 6933$, $P < 0.05$).

Most captured individuals were in the 93- to 100-mm-CL size class ($n = 1608$). In general, females were smaller than males, with females more frequently in samples in the smaller size classes (<100 mm), whereas males were more frequently found in the larger size classes (Fig. 2).

Linear regression analysis revealed negative allometric growth ($b < 1$) for TL v. CL and for TL v. AL relationships (i.e. CL and AL increase at a slower rate than TL). In contrast, the CL v. AL relationship reflected positive allometry ($b > 1$), with the rate of increase in AL greater than that of CL (Table 2).

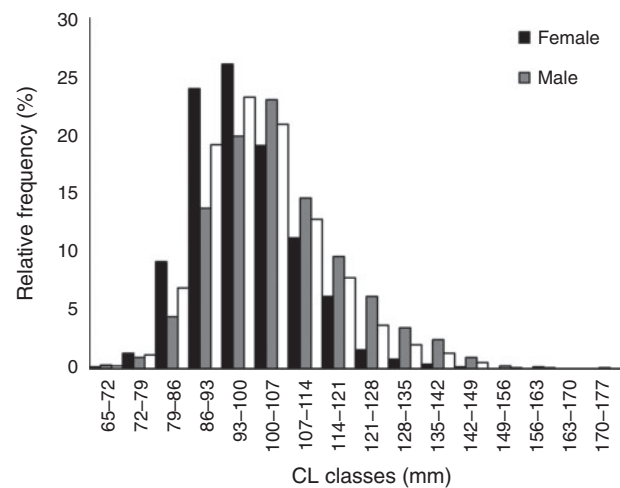
**Fig. 2.** Size–frequency distributions of cephalothorax length (CL) classes for *Panulirus meripurpuratus* collected off the Amazon coast in 2001 and 2002.

Fig. 3 and 4 show the von Bertalanffy growth parameters and growth performance index for the total (pooled dataset), as well as for males and females separately. Longevity for the entire dataset was 21.37 years.

Off the Amazon coast, the youngest age at which *P. meripurpuratus* are caught is ~3 years (~70 mm CL), with most individuals being caught between the ages of 5 and 7 years (CL 86–114 mm; Fig. 5).

Using k and L_{∞} values for the pooled dataset, the estimated total (Z), natural (M) and fishery (F) mortality was 0.93, 0.29 and 0.64 year⁻¹ respectively. These values indicate that, for 2001 and 2002, the exploitation rate of *P. meripurpuratus* caught off the Amazon coast was 0.69 year⁻¹, with a mean CL of 90.69 mm at first capture (CL_{50%}; i.e. 4–5 years old). During the study period, the exploitation rate of the lobster fishery, E , was 0.69, close to the maximum sustainable value of $E_{\max} = 0.83$ (Fig. 6).

Discussion

Lobsters are long-lived animals with low growth rates and longevity >35 years, although they reach ~90% of their L_{∞} value in 15 years (Instituto Brasileiro do Meio Ambiente e dos Recursos Naturais Renováveis 2008). Based on the sample length structure and growth parameter estimates obtained in this study, most individuals captured are between 86 and 107 mm CL. This corresponds to individuals ranging from 5 to 7 years of

Table 2. Regression equations for total length (TL) v. cephalothorax length (CL), TL v. abdomen length (AL) and CL v. AL for all *Panulirus meripurpuratus* (pooled dataset) collected off the Amazon coast in 2001 and 2002, and for males and females separately, as well as coefficients of determination (R^2) and F -test values for each regression

TL, CL and AL were measured in millimetres

Group	Number of individuals	Regression equation ($y = bx + a$)	R^2	F	P -value
TL v. CL					
Pooled dataset	6933	$CL = 0.4176 \times TL - 13.928$	0.86	208 784.9	<0.05
Male	3236	$CL = 0.4556 \times TL - 20.773$	0.93	84 363.45	<0.05
Female	3697	$CL = 0.3797 \times TL - 6.6364$	0.93	130 271.7	<0.05
TL v. AL					
Pooled dataset	6933	$AL = 0.5924 \times TL + 14.917$	0.82	54 845.23	<0.05
Male	3236	$AL = 0.5547 \times TL + 21.305$	0.86	26 784.29	<0.05
Female	3697	$AL = 0.6299 \times TL + 8.0726$	0.85	28 735.64	<0.05
CL v. AL					
Pooled dataset	6933	$AL = 1.1501 \times CL + 61.776$	0.62	54 081.02	<0.05
Male	3236	$AL = 1.1347 \times CL + 55.24$	0.80	29 598.32	<0.05
Female	3697	$AL = 1.5669 \times CL + 28.089$	0.81	54 081.02	<0.05

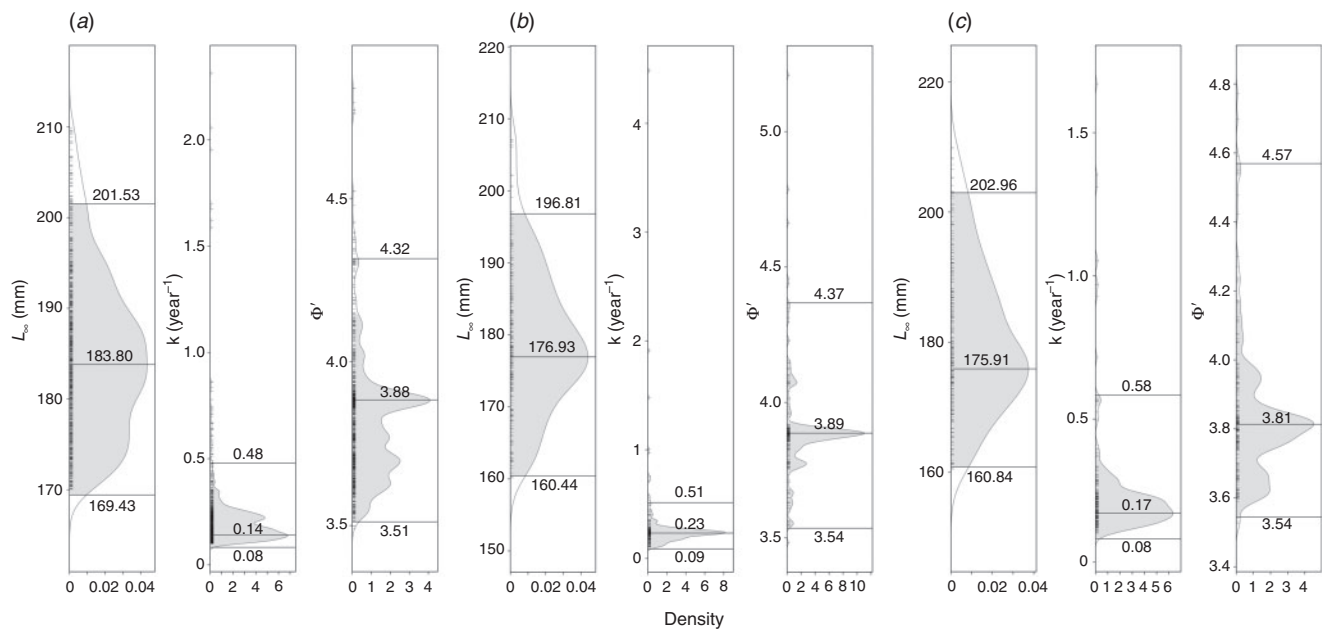


Fig. 3. Estimates of asymptotic length (L_{∞}), the growth constant (k) and the growth performance index (Φ') with confidence intervals for (a) all *Panulirus meripurpuratus* collected off the Amazon coast in 2001 and 2002, and for (b) males and (c) females separately.

age. The size at first capture (L_{50}) was estimated at 90.32 mm (4–5 years). The spiny lobster reaches its highest reproductive potential at ~ 4 years of age (Ivo and Gesteira 1986; Fonteles-Filho 2005), indicating that the fishery was predominantly exploiting individuals who have had at least one reproductive event. Silva *et al.* (2008) estimate that female spiny lobsters in northern Brazil reach their first sexual maturity at a CL of 69.20 mm and ~ 3 years of age.

Population parameters reported for spiny lobster studies in other regions differ from those obtained in the present study. Longevity > 37 years was estimated in north-eastern Brazil, and different estimates of L_{∞} and k have been reported (Table 3). These differences could be due to differences in environmental

conditions or sex ratios, sampling period, temperature, capture depth or even distinct fishery pressures (de León *et al.* 1994; Briones-Fourzán and Lozano-Alvarez 2002).

It is known that the intensity of fishery pressure affects the population size structure, reducing L_{∞} and accelerating k (Spare and Venema 1997). These changes represent a population-level response to a reduction in intraspecific competition resulting from the loss of larger individuals (Gulland 1956; Phillips *et al.* 1992; Leocádio and Cruz 2008). In this context, the pooled dataset for lobster stocks along the Amazon coast yields lower k values than those recorded in north-eastern Brazil and the Caribbean, given that the lobster fisheries in these latter regions have a much longer history (Ivo and Pereira 1996; Leocádio and

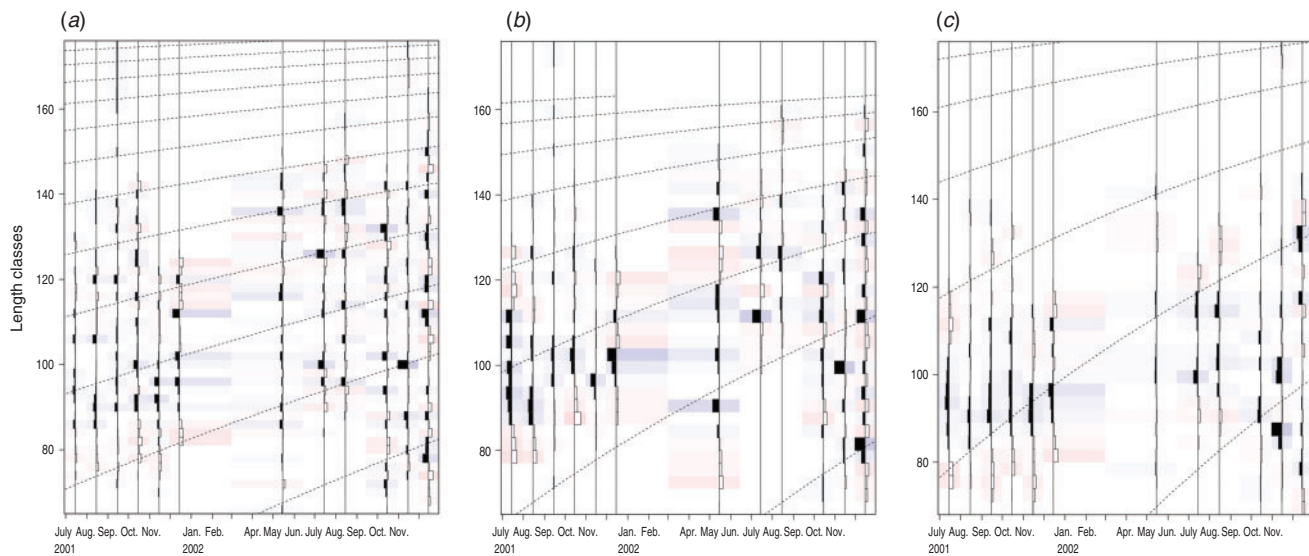


Fig. 4. Growth curves (dotted lines) based on cephalothorax length–frequency relationships obtained using the ELEFAN_GA model with bootstrapping for (a) all *Panulirus meripurpuratus* collected off the Amazon coast in 2001 and 2002, and for (b) males and (c) females separately. Black bars indicate positive peaks, whereas open bars indicate negative peaks. Shading refers to the difference between moving averages.

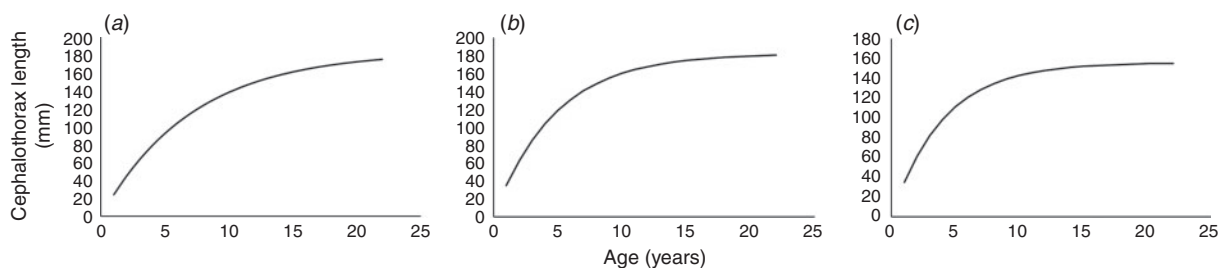


Fig. 5. Age–cephalothorax length key for (a) all *Panulirus meripurpuratus* collected off the Amazon coast in 2001 and 2002, and for (b) males and (c) females separately.

Cruz 2008) and their stocks are currently considered intensely overexploited (Cruz *et al.* 2013a).

The morphometric relationships recorded in this study confirmed a pattern of sexual dimorphism (Silva *et al.* 1994). The longer AL in females can be interpreted as an evolutionary adaptation to facilitate egg incubation during the reproductive period (Paiva 1960; Silva *et al.* 1994). This justifies the calculation of growth curves and age–length keys separately for the two sexes. However, the application of stock assessment models requires a single set of population parameters, because effective management measures cannot be applied differently to the two sexes considering that harvesting does not discriminate between the sexes.

The exploitation rate recorded for spiny lobster in the present study was lower than, but close to, the maximum sustainable exploitation rate. This indicates that the lobster stocks along the Amazon coast were not overexploited at the time of the data collection for this study and that any increase in fishery effort would result in overfishing.

Because of the absence of more recent data, it is important to note that the data analysed here are now outdated. In this

context, it is likely that the fishery effort has increased in recent times as a result of the failure to enforce controls and management policies. For this reason, an increase in fishing effort would not be recommendable nowadays. Increasing effort is a typical feature of fisheries. Particularly in developing countries, fishing effort has likely increased in over the years to levels of overexploitation, followed by a decline in catches (Stobutzki *et al.* 2006; Nomura 2010). Even during the study period there was a tendency for production to decrease over time, which may reflect either excessive effort or even overfishing (Pauly *et al.* 2002; Chávez 2009).

In the US, Caribbean and north-eastern Brazil there has been a decline in lobster catch volumes, reflecting the overexploitation of stocks (Chávez 2009; Ehrhardt *et al.* 2011; Andrade 2015). In Cuba, the production of lobster has decreased 80% over the past 30 years as a result of a combination of increasingly intense fishery pressure and habitat loss (Alzugaray Martínez and Puga Millán 2012). These changes have been associated with a high value of the growth constant (k) and reduced asymptotic length (L_{∞}) (Arce and de León 2001), as well as less intense recruitment and higher mortality rates in recently

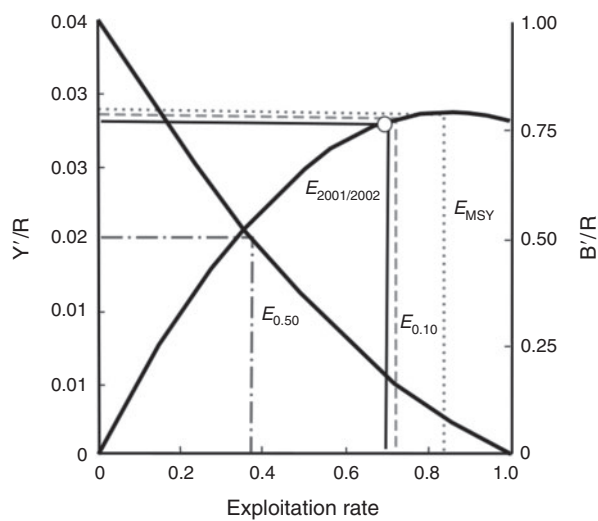


Fig. 6. Relative biomass per recruit (B/R , g) and yield per recruit (Y/R , g year⁻¹), based on the model of [Beverton and Holt \(1957\)](#) for the *Panulirus meripurpurus* specimens collected off the Amazon coast in 2001 and 2002. $E_{2001-2002}$, exploitation ratio (E) found in the present study (2001–02), as indicated by the open circle; E_{MSY} , maximum sustainable yield exploitation ratio; $E_{0.10}$, exploitation rate when the slope of the yield curve is 10% of the initial curve; $E_{0.50}$, exploitation rate where virgin biomass is 50%.

Table 3. Values of von Bertalanffy growth curve parameters for the spiny lobster in different studies in different regions
 L_{∞} , asymptotic length; k , growth constant

Region	L_{∞} (mm)	k (year ⁻¹)	Reference
Brazil	257	0.32	Ivo and Pereira (1996)
Mexico	142	0.22	Arce (1990)
Cuba	174	0.16	Buesa (1972)
Cuba	228	0.30	Báez et al. (1991)
Cuba	189	0.27	de León et al. (2005)

recruited individuals ([Alzugaray et al. 2018](#)). In addition to the intensity of the fishery, changes in population dynamics have been interpreted as a consequence of the loss of habitat, the lack of nutrients, an increase in extreme events and rising sea temperatures, which also affect lobster growth ([de León et al. 2005](#); [Puga et al. 2013](#)).

The Amazon region is experiencing increasingly intense environmental impacts, which may affect lobster populations. Human activities, including petroleum and heavy metal pollution, deforestation and the construction of dams in both central drainage basins and coastal areas, are changing the flow of nutrients to the ocean ([Lima et al. 2015](#); [Latrubesse et al. 2017](#); [Castello et al. 2018](#)). A large reef system discovered recently on the Amazon continental shelf that is a good environment for lobster recruitment and feeding may be threatened by fishing and other anthropogenic actions ([Moura et al. 2016](#)).

Current models of climate change indicate that the volume of rainfall in the eastern Amazon basin will decline over the next few decades ([Betts et al. 2008](#)) and that the frequency of extreme

climatic events will progressively increase ([Zeng et al. 2008](#); [Marengo et al. 2013](#)). All these processes will likely affect the productivity of the waters in which the lobsters are found, and require rigorous controls and regulation, which, at present, are non-existent in Brazil.

More than a decade after the collection of the biometric data analysed in the present study, no data are available on the current status of the lobster stocks off the Amazon coast. It seems reasonable to assume that the exploitation status of these stocks has worsened over the past few years, which could be confirmed by the collection of new data on productivity and fishery efforts, as well as the data needed to calculate new population parameter estimates. Nevertheless, it is highly recommended that a systematic program of continuous monitoring is implemented to compile data on biometric patterns and fishery parameters. Without reliable data on these parameters, it would be advisable to adopt the findings of the present study for the development of effective management policies for the lobster fisheries off the Amazon coast, which should nevertheless be based on as cautious an interpretation of the data as possible.

The Brazilian government has long attempted to establish adequate policies for the regulation of the country's lobster fisheries to guarantee the sustainability of the resource and for the conservation of the spiny lobster ([Silva and Fonteles-Filho 2011](#); [Cavalcante and Furtado-Neto 2012](#)). However, this approach has continuously changed due to a lack of success, and there has been little effective monitoring to determine whether policies have been implemented adequately and, in particular, whether stocks have recovered. The closed season is regulated by federal Normative Instruction 206 of 2008 ([Instituto Brasileiro do Meio Ambiente e dos Recursos Naturais Renováveis 2008](#)), which prohibits lobster fishing between 1 December and 31 May and specifies a minimum abdomen size of 130 mm for the first capture, which corresponds to a CL of 75 mm. The legislation also prohibits the use of bottom-set gill nets and commercial diving in natural and artificial refuges, in addition to restricting fishery activity within a given distance of the coast. There was also an attempt to implement fishery quotas, although this measure was unsuccessful and was discontinued.

In general, these attempts regulating and controlling Brazilian lobster fisheries have been mostly ineffective, and the public authorities themselves recognise that the legislation is widely ignored and that there is a considerable illegal catch. This is probably due to the lack of resources for the effective implementation of control measures ([Cruz et al. 2013a](#)), together with the autocratic approach to the management of fisheries in Brazil in general ([Isaac 2006](#)). Overall, this situation may lead to the loss of sustainability and the exhaustion of stocks, with severe implications for the region's economy and those relying on this activity for their livelihood. New studies are needed to evaluate the levels of illegal harvesting and to update the estimates of fishery effort. Continuous monitoring of catches and fishery effort, together with the regular collection of biometric data, will be fundamental to the reliable, updated evaluation of fishery production, permitting optimisation of management measures and the effective administration of the lobster stocks along the Amazon coast.

Conflicts of interest

The authors declare that they have no conflicts of interest.

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References

- Alzugaray, R., Puga, R., Piñeiro, R., de León, M. E., Cobas, L. S., and Morales, O. (2018). The Caribbean spiny lobster (*Panulirus argus*) fishery in Cuba: current status, illegal fishing, and environmental variability. *Bulletin of Marine Science* **94**(2), 393–408. doi:10.5343/BMS.2016.1126
- Alzugaray Martínez, R., and Puga Millán, R. (2012). Comparación entre dos modelos estructurados por edades, aplicados a la pesquería, *Panulirus argus* (Latreille, 1804), en la región suroriental de Cuba. *Revista de Ciencias Marinas y Costeras* **4**, 131–143. doi:10.15359/REVMAR.4.9
- Andrade, H. (2015). Stock assessment of the red spiny lobster (*Panulirus argus*) caught in the tropical southwestern Atlantic. *Latin American Journal of Aquatic Research* **43**(1), 201–214. doi:10.3856/VOL43-ISSUE1-FULLTEXT-17
- Andrades, R., Machado, F. S., Reis-Filho, J. A., Macieira, R. M., and Giarrizzo, T. (2018). Intertidal biogeographic subprovinces: local and regional factors shaping fish assemblages. *Frontiers in Marine Science* **5**, 412. doi:10.3389/FMARS.2018.00412
- Arce, A. M. (1990). Estimación del crecimiento de langosta *Panulirus argus* (Latreille) con metodologías convencionales y a través de funciones de singularidad. M.Sc. Thesis, Instituto Politécnico Nacional, Centro de Investigaciones de Estudios Avanzados Unidad Mérida, Mérida, Yucatán, Mexico.
- Arce, A. M., and de León, M. E. (2001). Biology. In 'Report on the FAO/DANIDA/CFRAMP/WECAFC Regional Workshops on the Assessment of the Caribbean Spiny Lobster (*Panulirus argus*)'. (Eds P. Medley and S. Venema.) FAO Fishery Report number 619, pp. 17–25 (Food and Agriculture Organization of the United Nations: Rome, Italy.)
- Báez, M., Díaz, E., Brito, R., and Cruz, R. (1991). Edad y crecimiento de la langosta en la plataforma suroccidental de Cuba. *Revista de Investigaciones Marinas* **12**, 193–201.
- Baeza, J. A., Simpson, L., Ambrosio, L. J., Mora, N., Guéron, R., and Childress, M. J. (2016). Active parental care, reproductive performance, and a novel egg predator affecting reproductive investment in the Caribbean spiny lobster *Panulirus argus*. *BMC Zoology* **1**, 6. doi:10.1186/S40850-016-0006-6
- Barthelm, R. B., and Schwassmann, H. O. (1994). The Amazon River influence over the seasonal displacement of the salty wedges in Tocantins Estuary, Brazil, 1983–1985. *Boletim do Museu Paraense Emílio Goeldi* **10**(1), 119–130.
- Betts, R. A., Sanderson, M., and Woodward, S. (2008). Effects of large-scale Amazon forest degradation on climate and air quality through fluxes of carbon dioxide, water, energy, mineral dust and isoprene. *Philosophical Transactions of The Royal Society* **363**(1498), 1873–1880. doi:10.1098/RSTB.2007.0027
- Beverton, R. J. H., and Holt, S. J. (1957). On the dynamics of exploited fish populations. Fisheries Investigations Series II, volume 19. Ministry of Agriculture, Fisheries, and Food, Her Majesty's Stationery Office, London, UK.
- Bowen, B. K. (1980). Spiny lobster management. In 'The Biology and Management of Lobsters'. (Eds J. S. Cobb and B. F. Phillips.) Vol. 2, pp. 243–264. (Academic Press: New York, NY, USA.)
- Briones-Fourzán, P., and Lozano-Álvarez, E. (2002) Shallow-water benthic decapod crustaceans of Chankanaab Park, Cozumel Island, México. In 'Modern Approaches to the Study of Crustaceans'. (Eds E. Escobar-Briones and F. Álvarez.) pp. 197–204. (Kluwer: Amsterdam, Netherlands.)
- Buesa, R. (1972). La langosta espinosa *Panulirus argus*: su pesca y biología en aguas cubanas. INP/CIP, Cuba. *Reunión Balance de Trabajo* **3**(1), 29–78.
- Butler, C. B., and Matthews, T. R. (2015). Effects of ghost fishing lobster traps in the Florida Keys. *ICES Journal of Marine Science* **72**(1), i185–i198. doi:10.1093/ICESJMS/FSU238
- Castello, L., Hess, L. L., Thapa, R., McGrath, D. G., Arantes, C. C., Renó, V. F., and Isaac, V. J. (2018). Fishery yields vary with land cover on the Amazon River floodplain. *Fish and Fisheries* **19**, 431–440. doi:10.1111/FAF.12261
- Cavalcante, P. P. L., and Furtado-Neto, M. A. A. (2012). Implementação de quotas individuais transferíveis e do desembarque compulsório de lagosta viva como estratégia de ordenamento da pesca. *Arquivos de Ciências do Mar* **45**(2), 49–59.
- Chávez, E. (2009). Potential production of the Caribbean spiny lobster (Decapoda, Palinura) fisheries. *Crustaceana* **82**(11), 1393–1412. doi:10.1163/001121609X12481627024373
- Conrad, K., and Danoff-Burg, J. (2011). Lobster houses as a sustainable fishing alternative. *Consilience* **6**, 48–62. doi:10.7916/CONSILIENCE.V016.4569
- Cruz, R., Lalana, R., Báez-Hidalgo, M., and Adriano, R. (2007). Gregarious behaviour of juveniles of the spiny lobster, *Panulirus argus* (Latreille, 1804) in artificial shelters. *Crustaceana* **80**(5), 577–595. doi:10.1163/156854007780765597
- Cruz, R., Silva, K. C. A., Neves, S. D. S., and Cintra, I. H. A. (2013a). Impact of lobster size on catches and prediction of commercial spiny lobster landings in Brazil. *Crustaceana* **86**(10), 1274–1290. doi:10.1163/15685403-00003230
- Cruz, R., Silva, K. C. A., and Cintra, I. H. A. (2013b). Assessment of wild spiny lobster stocks on the Brazilian continental shelf. *Crustaceana* **86**(3), 336–356. doi:10.1163/15685403-00003173
- Cruz, R., Silva, K. A., Santana, J. M., Gaeta, J. C., and Cintra, I. A. (2014). Reproductive potential and stock recruitment of the Caribbean and Brazilian metapopulations of the spiny lobster, *Panulirus argus* (Latreille, 1804). *Crustaceana* **87**(11–12), 1315–1337. doi:10.1163/15685403-00003347
- Cruz, R., Teixeira, C. P., Menezes, M. B., Santana, J. M., Neto, T. M., Gaeta, J. C., De Freitas, P. P., Silva, K. A., and Cintra, I. A. (2015). Large-scale oceanic circulation and larval recruitment of the spiny lobster *Panulirus argus* (Latreille, 1804). *Crustaceana* **88**(3), 298–323. doi:10.1163/15685403-00003411
- Dall'occo, P. L. (2010). 'Taxonomia e Distribuição das Lagostas (Crustacea: Decapoda: Achelata e Polychelida) no Oceano Atlântico.' (Universidade Estadual Paulista, Rio Claro, São Paulo, Brazil.)
- de León, M. E., Puga, R., and Cruz, R. (1994). Estimación de los parámetros de crecimiento de *Panulirus argus* em el Golfo de Batabanó, Cuba. *Revista Cubana Investigaciones Pesqueras* **18**(1), 9–12.
- de León, M. E., López Martínez, J., Lluch Cota, D., Hernández Vázquez, S., and Puga, R. (2005). Decadal variability in growth of the Caribbean spiny lobster *Panulirus argus* (Decapoda: Paniluridae) in Cuban waters. *Revista de Biología Tropical* **53**(3–4), 475–486. doi:10.15517/RBT.V53I3-4.14616

- Di Paolo, I. F., Gouveia, N. A., Neto, L. C. F., Paes, E. T., Vijaykumar, N. L., and Santana, Á. L. (2019). Data mining of spatio-temporal variability of chlorophyll-*a* concentrations in a portion of the western Atlantic with low performance hardware. *Journal of Software Engineering and Applications* **12**(05), 149. doi:10.4236/JSEA.2019.125010
- Ehrhardt, N. M. (2008). Estimating growth of the Florida spiny lobster, *Panulirus argus*, from molt frequency and size increment data derived from tag and recapture experiments. *Fisheries Research* **93**(3), 332–337. doi:10.1016/J.FISHRES.2008.06.008
- Ehrhardt, N. M., Puga, R., and Butler, M. J. (2011). Implications of the ecosystem approach to fisheries management in large ecosystems. The case of the Caribbean spiny lobster. In 'Towards Marine Ecosystem-Based Management in the Wider Caribbean'. (Eds L. Fanning, R. Mahon, and P. McConney.) pp. 157–175. (Amsterdam University Press: Amsterdam, Netherlands.)
- Fonteles-Filho, A. A. (1979). Biologia pesqueira e dinâmica populacional da Lagosta *Panulirus laeviscauda* (Latreille), no Nordeste setentrional do Brasil. *Arquivos de Ciências do Mar* **19**(1–2), 1–43. doi:10.32360/ACMAR.V19I1-2.31601
- Fonteles-Filho, A. A. (1986). Influência do recrutamento e da pluviosidade sobre a abundância da lagosta *Panulirus argus* (Latreille) e *Panulirus laeviscauda* (Latreille) (Crustacea: Palinuridae), no Nordeste do Brasil. *Arquivos de Ciências do Mar* **25**, 13–31. doi:10.32360/ACMAR.V25I1-2.31453
- Fonteles-Filho, A. A. (1997). Spatial distribution of the lobster species *Panulirus argus* and *P. laeviscauda* in northern and northeastern Brazil in relation to the distribution of fishing effort. *Ciência e Cultura* **49**(3), 172–176.
- Fonteles-Filho, A. A. (2005). Síntese sobre distribuição, abundância, potencial pesqueiro e biologia da lagosta-vermelha *Panulirus argus* (Latreille) e a lagosta verde *Panulirus laeviscauda* (Latreille) do nordeste do Brasil. Avaliação do Potencial Sustentável de Recursos Vivos na Zona Econômica Exclusiva (REVIZEE), Ministério do Meio Ambiente (MMA), Brasília, Brazil.
- Food and Agriculture Organization of the United Nations (2003). Fisheries Report/FAO Informe de Pesca. Number 715, FAO, Rome, Italy.
- Food and Agriculture Organization of the United Nations (2019). 'FAO Yearbook. Fishery and Aquaculture Statistics 2017.' (FAO: Rome, Italy.)
- George, R. W. (2005). Review: Evolution of life cycles, including migration, in spiny lobsters (Palinuridae). *New Zealand Journal of Marine and Freshwater Research* **39**(3), 503–514. doi:10.1080/00288330.2005.9517329
- Giraldes, B. W., and Smyth, D. M. (2016). Recognizing *Panulirus meripurpuratus* sp. nov. (Decapoda: Palinuridae) in Brazil – systematic and biogeographic overview of *Panulirus* species in the Atlantic Ocean. *Zootaxa* **4107**, 353. doi:10.11646/ZOOTAXA.4107.3.4
- Gnanalingam, G., Butler, M. J., IV, Matthew, T. R., Hutchinson, E., and Kilada, R. (2019). Directly ageing the Caribbean spiny lobster, *Panulirus argus*, with validated band counts from gastric mill ossicles. *ICES Journal of Marine Science* **76**, 442–451. doi:10.1093/ICESJMS/FSY177
- Gouveia, N. A., Gherardi, D. F. M., and Aragão, L. E. O. C. (2019a). The role of the Amazon River plume on the intensification of the hydrological cycle. *Geophysical Research Letters* **46**(21), 12221–12229. doi:10.1029/2019GL084302
- Gouveia, N. A., Gherardi, D. F. M., Wagner, F. H., Paes, E. T., Coles, V. J., and Aragão, L. E. O. C. (2019b). The salinity structure of the Amazon River plume drives spatiotemporal variation of oceanic primary productivity. *Journal of Geophysical Research – Biogeosciences* **124**(1), 147–165. doi:10.1029/2018JG004665
- Gulland, J. A. (1955). On the estimation of population parameters from marked members. *Biometrika* **42**(1–2), 269–270. doi:10.1093/BIOMET/42.1-2.269
- Gulland, J. A. (1956). On the fishing effort in English demersal trawl fisheries. *Fisheries Investigations Series II. Ministry of Agriculture, Fisheries and Food UK* **20**, 1–41.
- Instituto Brasileiro do Meio Ambiente e dos Recursos Naturais Renováveis (2005). 'Estatística da Pesca 2004: grandes regiões e unidades da Federação.' (IBAMA and Ministério do Meio Ambiente, Brasília, Brazil.)
- Instituto Brasileiro do Meio Ambiente e dos Recursos Naturais Renováveis (2007). 'Estatística da Pesca 2007: grandes regiões e unidades da Federação'. (IBAMA and Ministério do Meio Ambiente, Brasília, Brazil.)
- Instituto Brasileiro do Meio Ambiente e dos Recursos Naturais Renováveis (2008). 'Plano de gestão para o uso sustentável de Lagostas no Brasil: *Panulirus argus* (Latreille, 1804) e *Panulirus laeviscauda* (Latreille, 1817).' (IBAMA and Ministério do Meio Ambiente, Brasília, Brazil.)
- Isaac, V. J. (2006). Exploração e manejo dos recursos pesqueiros do litoral amazônico: um desafio para o futuro. *Ciência e Cultura* **58**(3), 33–36.
- Isaac, V. J., and Ferrari, S. F. (2017). Assessment and management of the North Brazil Shelf Large Marine Ecosystem. *Environmental Development* **22**, 97–110. doi:10.1016/J.ENVDEV.2016.11.004
- Isaac, V. J., Santo, R. V., Bentes, B., Frédou, F. L., Mourão, K. R., and Frédou, T. (2009). An interdisciplinary evaluation of fishery production systems off the state of Pará in North Brazil. *Journal of Applied Ichthyology* **25**, 244–255. doi:10.1111/J.1439-0426.2009.01274.X
- Ivo, C. T. C., and Gesteira, T. C. V. (1986). Potencial reprodutivo das lagostas *Panulirus argus* e *Panulirus laeviscauda* (Latreille) (Crustacea: Palinuridae), no Nordeste do Brasil. *Arquivos de Ciências do Mar* **25**, 1–12.
- Ivo, C. T. C., and Pereira, J. A. (1996). Sinopse das principais observações sobre as lagostas *Panulirus argus* (Latreille) e *Panulirus laeviscauda* (Latreille), capturadas em águas costeiras do Brasil, entre os Estados do Amapá e do Espírito Santo. *Boletim Técnico Científico CEPENE* **4**(1), 7–94.
- Kilada, B., Sainte-Marie, B., Rochette, R., Davis, N., Vanier, C., and Campana, S. (2012). Direct determination of age in shrimps, crabs, and lobsters. *Canadian Journal of Fisheries and Aquatic Sciences* **69**, 1728–1733. doi:10.1139/CJFAS-2012-0254
- King, M. (2007). 'Fisheries Biology: Assessment and Management', 2nd edn. (New Books: Osney Mead, UK.)
- Latrubesse, E. M., Arima, E. Y., Dunne, T., Park, E., Baker, V. R., d'Horta, F. M., Wight, C., Wittmann, F., Zuanon, J., Baker, P. A., Ribas, C. C., Norgaard, R. B., Filizola, N., Ansar, A., Flyvbjerg, B., and Stevaux, J. C. (2017). Damming the rivers of the Amazon basin. *Nature* **546**(7658), 363–369. doi:10.1038/NATURE22333
- Leocádio, A. M., and Cruz, R. (2008). Growth parameters of the spiny lobster (*Panulirus argus*) in the great Caribbean: a review. *Revista de Investigaciones Marinas* **29**(3), 239–248.
- Lima, D. P., Santos, C., Silva, R. S., Yoshioka, E. T. O., and Bezerra, R. M. (2015). Contaminação por metais pesados em peixes e água da bacia do rio Cassiporé, Estado do Amapá, Brasil. *Acta Amazonica* **45**(4), 405–414. doi:10.1590/1809-4392201403995
- Lima, F. A., Martinelli-Lemos, J. M., Silva, K. A., Klautau, A. M., and Cintra, I. A. (2018). Population structure and fecundity of *Scyllarides delfosi* Holthuis, 1960 (Scyllaridae) on the Amazon continental shelf. *Crustaceana* **91**(9), 1027–1037. doi:10.1163/15685403-00003783
- Marengo, J. A., Alves, L. M., Soares, W. R., Rodriguez, D. A., Camargo, H., Riveros, M. P., and Pablo, A. D. (2013). Two contrasting severe seasonal extremes in tropical South America in 2012: flood in Amazonia and drought in northeast Brazil. *Journal of Climate* **26**, 9137–9154. doi:10.1175/JCLI-D-12-00642.1
- Medeiros, P. M., Seidel, M., Ward, N. D., Carpenter, E. J., Gomes, H. R., Niggemann, J., Krusche, A. V., Richey, J. E., Yager, P. L., and Dittmar, T. (2015). Fate of the Amazon River dissolved organic matter in the tropical Atlantic Ocean. *Global Biogeochemical Cycles* **29**(5), 677–690. doi:10.1002/2015GB005115
- Ministério da Pesca e Aquicultura (2011). Boletim Estatístico da pesca e aquicultura, Brasil. MPA, Brasília, Brazil.

- Moura, R. L., Amado-Filho, G. M., Moraes, F. C., Brasileiro, P. S., Salomon, P. S., Mahiques, M. M., Bastos, A. C., Almeida, M. G., Silva, J. M., Araujo, B. F., Brito, F. P., Rangel, T. P., Oliveira, B. C. V., Bahia, R. G., Paranhos, R. P., Dias, R. J. S., Siegle, E., Figueiredo, A. G., Pereira, R. C., Leal, C. V., Hajdu, E., Asp, N. E., Gregoracci, G. B., Neumann-Leitão, S., Yager, P. L., Francini-Filho, R. B., Fróes, A., Campeão, M., Silva, B. S., Moreira, A. P. B., Oliveira, L., Soares, A. C., Araujo, L., Oliveira, N. L., Teixeira, J. B., Valle, R. A. B., Thompson, C. C., Rezende, C. E., and Thompson, F. L. (2016). An extensive reef system at the Amazon River mouth. *Science Advances* **2**(4), e1501252. doi:10.1126/SCIADV.1501252
- Nomura, I. (2010). O futuro da pesca e da aquicultura marinha no mundo. *Ciência e Cultura* **62**(3), 28–32.
- Oliveira, V. S., Andrade, H. A., Hazin, F. H. V., and Vieira, A. C. (2015). Pesca da lagosta com covos na costa central de Pernambuco: taxas de captura da lagosta e fauna acompanhante. *Boletim do Instituto de Pesca* **41**(2), 373–385. doi:10.20950/1678-2305.2017V43N2P194
- Paiva, M. P. (1960). Dimorfismo sexual observado em relações de peso e comprimento da lagosta *Panulirus argus* (Latreille). *Revista Brasileira de Biologia* **20**(1), 51–62.
- Paiva, M. P. (1997). 'Recursos pesqueiros estuarinos e marinhos do Brasil.' (Editora da Universidade Federal do Ceará: Fortaleza, Brazil.)
- Pauly, D. (1980). On the interrelationships between natural mortality, growth parameters, and mean environmental temperature in 175 fish stocks. *ICES Journal of Marine Science* **39**(2), 175–192. doi:10.1093/ICESJMS/39.2.175
- Pauly, D., Christensen, V., Guénette, S., Pitcher, T. J. U., Sumaila, R., Walters, C. J., Watson, R., and Zeller, D. (2002). Towards sustainability in world fisheries. *Nature* **418**, 689–695. doi:10.1038/NATURE01017
- Penn, J. W., Caputi, N., and Lestang, S. (2015). A review of lobster fishery management: the Western Australian fishery for *Panulirus cygnus*, a case study in the development and implementation of input and output-based management systems. *ICES Journal of Marine Science* **72**(1), i22–i34. doi:10.1093/ICESJMS/FSV057
- Phillips, B. F., Palmer, M. J., Cruz, R., and Trendall, J. T. (1992). Estimating growth of the spiny lobsters *Panulirus cygnus*, *P. argus* and *P. ornatus*. *Marine and Freshwater Research* **43**, 1177–1188. doi:10.1071/MF9921177
- Phillips, B. F., Melville-Smith, R., Kay, M. C., and Vega-Velázquez, A. (2013). *Panulirus* species. In 'Lobsters: Biology, Management, Aquaculture and Fisheries'. (Ed. B. F. Phillips.) pp. 289–325. (Wiley–Blackwell: Oxford, UK.)
- Porto, V. M. S., Cintra, I. H. A., and Silva, K. C. A. (2005). Sobre a pesca da lagosta-vermelha, *Panulirus argus* (Latreille, 1804), na costa norte do Brasil. *Boletim Técnico-Científico do CEPNOR* **5**(1), 83–92. doi:10.17080/1676-5664/BTCC.V5N1P83-92
- Puga, R., Piñeiro, R., Alzugaray, R., Cobas, L. S., de León, M. E., and Morales, O. (2013). Integrating anthropogenic and climatic factors in the assessment of the Caribbean spiny lobster (*Panulirus argus*) in Cuba: implications for fishery management. *International Journal of Marine Science* **3**(6), 36–45. doi:10.5376/IJMS.2013.03.0006
- Ricker, W. E. (1975). Computation and interpretation of biological statistics of fish populations. Fisheries and Marine Service, Department of the Environment, Ottawa, ON, Canada.
- Santana, W., Ivo, C. T. C., Neto, J. D., Duarte, L. F. A., Pinheiro, M. A. A., Boos, H., Pinheiro, A., Almeida, A. O., Hernáez, P., and Coelho, P. A. (2016). Avaliação das lagostas-de-espinho (Decapoda: Palinuridae). In 'Livro Vermelho dos Crustáceos do Brasil: Avaliação 2010–2014'. (Eds M. A. A. Pinheiro and H. Boos.) pp. 268–283. (Sociedade Brasileira de Carcinologia: Porto Alegre, Rio Grande do Sul, Brazil.) Available at <http://www.crusta.com.br/biblio/03.e-books/02-Livro-Vermelho-dos-Crustaceos-do-Brasil-Avaliacao-2010-2014.pdf> [Verified 15 May 2020].
- Santos, M. C. F., and Freitas, A. E. T. S. (2002). Estudo sobre a lagosta sapata (*Scyllarides brasiliensis*) Rathbun, 1906 (Crustacea: Decapoda: Scyllaridae) no litoral dos estados de Pernambuco e Alagoas – Brasil. *Boletim Técnico-Científico do CEPENE* **10**(1), 123–143.
- Santos, F. J. S., Silva, K. C. A., Bentes, B., Pereira, M. E. G. S., Klautau, A. G. C. M., and Cintra, I. A. (2019). A pesca de lagostas na Plataforma Continental Amazônica. *Arquivos de Ciências do Mar* **52**(2), 61–76.
- Sanyo Techno Marine (1998). Draft final report for the fishery resources study of the Amazon and Tocantins river mouth areas in the Federative Republic of Brazil. Sanyo Techno Marine, Tokyo, Japan.
- Schwaborn, R., Mildenberger, T. K., and Taylor, M. H. (2019). Assessing sources of uncertainty in length-based estimates of body growth in populations of fishes and macroinvertebrates with bootstrapped ELEFAN. *Ecological Modelling* **393**, 37–51. [Published erratum appears in *Ecological Modelling* **2019**, 394, 76–77, <https://doi.org/10.1016/j.ecolmodel.2019.01.003>]. doi:10.1016/J.ECOLMODEL.2018.12.001
- Silva, A. C., and Fonteles-Filho, A. A. (2011). 'Avaliação do Defeso Aplicado à Pesca da Lagosta no Nordeste do Brasil.' (Editora Expressão Gráfica: Fortaleza, Ceará, Brazil.)
- Silva, J. R. F., Gesteira, T. C. V., and Rocha, C. A. S. (1994). Relações morfométricas ligadas à reprodução da lagosta espinhosa, *Panulirus laeviscauda* (Latreille) do Estado do Ceará. *Boletim Técnico Científico CEPENE* **2**(1), 59–88.
- Silva, K. C. A., Cintra, I. H. A., Porto, M. R., and Viana, G. F. S. (2003). Lagostas capturadas durante pescarias experimentais para o programa REVIZEE/NORTE (Crustacea, Nephropoidea, Eryonoidea, Palinuroidea). *Boletim Técnico-Científico do CEPNOR* **3**(1), 21–35.
- Silva, A., Araujo, M., Medeiros, C., Silva, M., and Bourles, B. (2005). Seasonal changes in the mixed and barrier layers in the western Equatorial Atlantic. *Brazilian Journal of Oceanography* **53**(3–4), 83–98. doi:10.1590/S1679-87592005000200001
- Silva, K. C. A., Cintra, I. H. A., Porto, M. R., and Viana, G. F. S. (2007). Lagostas capturadas na plataforma continental do estado do Amapá (Crustacea, Nephropidae, Palinuroidea). *Boletim Técnico-Científico do CEPNOR* **7**, 173–184. doi:10.17080/1676-5664/BTCC.V7N1P173-184
- Silva, K. C. A., Botelho, K. B., Cintra, I. H. A., and Aragão, J. A. N. (2008). Parâmetros reprodutivos da lagosta-vermelha *Panulirus argus* (Latreille, 1804), na costa Norte do Brasil. *Boletim Técnico-Científico do CEPNOR* **8**, 27–39. doi:10.17080/1676-5664/BTCC.V8N1P27-39
- Silva, A. C., Santos, M. L. S., Araujo, M. C., and Bourlès, B. (2009). Observações hidrológicas e resultados de modelagem no espalhamento sazonal e espacial da pluma de água Amazônica. *Acta Amazonica* **39**(2), 361–369. doi:10.1590/S0044-59672009000200014
- Silva, K. A., Cruz, R., Cintra, I. A., and Abrunhosa, F. A. (2013). Structure and diversity of the lobster community on the Amazon continental shelf. *Crustaceana* **86**(9), 1084–1102. doi:10.1163/15685403-00003227
- Skewes, T. D., Pitcher, C. R., and Dennis, D. M. (1997). Growth of ornate rock lobsters, *Panulirus ornatus*, in Torres Strait, Australia. *Marine and Freshwater Research* **48**, 497–501. doi:10.1071/MF96063
- Spare, P., and Venema, S. (1997). Introduction to tropical fish stock assessment. Part 1 – manual. FAO Fisheries Technical Paper 306.1, Food and Agriculture Organization of the United Nations: Rome, Italy.
- Stobutzki, I. C., Silvestre, G. T., Abu Talib, A., Krongprom, A., Supongpan, M., Khemakorn, P., Armada, N., and Garces, L. R. (2006). Decline of demersal coastal fisheries resources in three developing Asian countries. *Fisheries Research* **78**, 130–142. doi:10.1016/J.FISHRES.2006.02.004
- Taylor, C. C. (1958). Cod growth and temperature. *Journal du Conseil - Conseil International pour l'Exploration de la Mer* **23**, 366–370. doi:10.1093/ICESJMS/23.3.366
- Tubino, R. A., Monteiro-Neto, C., Moraes, L. E. S., and Paes, E. T. (2007). Artisanal fisheries production in the coastal zone of Itaipu, Niterói, RJ, Brazil. *Brazilian Journal of Oceanography* **55**(3), 187–197. doi:10.1590/S1679-87592007000300003
- Vasconcelos, J. A., Vasconcelos, E. M., and Lins-Oliveira, J. E. (1994). Captura por unidade de esforço dos diferentes métodos de pesca (rede,

- mergulho e covo) empregados na pesca lagosteira do Rio Grande do Norte. *Boletim Técnico-Científico do CEPENE* **2**(1), 133–153.
- von Bertalanffy, L. (1938). A quantitative theory of organic growth (inquiries on growth laws. ii). *Human Biology* **10**(2), 181–213.
- Ward, T. J., and Phillips, B. F. (2013). Lobster ecolabelling. In ‘Lobsters: Biology, Management, Aquaculture and Fisheries’, 2nd edn. (Ed. B. F. Phillips.) pp. 139–185. (Wiley–Blackwell: Oxford, UK.)
- Zeng, N., Yoon, J. H., Marengo, J. A., Subramaniam, A., Nobre, C. A., Mariotti, A., and Neelin, J. D. (2008). Causes and impacts of the 2005 Amazon drought. *Environmental Research Letters* **3**, 014002. doi:[10.1088/1748-9326/3/1/014002](https://doi.org/10.1088/1748-9326/3/1/014002)
- Zhang, Y., and Yao, N. (2018). Estimating growth of Caribbean spiny lobster using mark–recapture data. *Marine and Coastal Fisheries* **10**, 481–492. doi:[10.1002/MCF2.10046](https://doi.org/10.1002/MCF2.10046)

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